

Quantifying War Damage from Satellite Imagery*

Nils Neuner^{†1}, Marcel Kalb^{‡1}, Mike Barth^{§1}, and Sam Känner^{¶1}

¹Department of Politics and Public Administration, University of Konstanz, Konstanz, Germany

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Lecturer: Dr. Giordano De Marzo



*Data and replication code are available at github.com/rudimedia/war_damage

[†]nils.neuner@uni-konstanz.de

[‡]marcel.julian.kalb@uni-konstanz.de

[§]mike.barth@uni-konstanz.de

[¶]sam.kaenner@uni-konstanz.de

1 Introduction

The goal of this study is to develop a tool for estimating building destruction following armed conflict and, more broadly, to provide information that can support policy analysis. Beyond measuring overall destruction, such a tool could be used to monitor damage to specific types of buildings, such as hospitals or schools. We focus on Gaza for both policy and practical reasons. The scale of destruction makes Gaza an important case for developing and evaluating satellite-based damage assessment methods, particularly in a context where independent access to the territory is severely restricted. There was also a practical reason for limiting the scope of the study to Gaza. Following an academic data request, we were granted access to PlanetScope imagery subject to a monthly quota sufficient to cover an area approximately the size of the Gaza Strip. We therefore focus our analysis on Gaza rather than attempting to extend it across multiple conflict zones.

2 Data strategy

Our data strategy requires both labels and features. The United Nations Satellite Centre (UNOSAT) provides the building-level damage labels, while Sentinel-1 and PlanetScope provide the satellite imagery used as model features.

Labels UNOSAT provides building-level damage assessments based on very-high-resolution commercial satellite imagery, including WorldView-2 and WorldView-3. We use assessments from 3 May, 6 July, and 6 September 2024. For the Sentinel-1 analysis, all three dates contribute observations to model development and evaluation. At each date, northern Gaza is used for training, intermediate latitude bands for stacking and validation, and southern Gaza for testing. Buildings therefore remain geographically separated across the development and test sets, while the model is trained jointly on observations from all three assessment dates. Reporting test performance separately for May, July, and September allows us to assess whether the learned relationship between Sentinel-1 imagery and building damage remains stable as damage prevalence, spatial patterns, and imagery conditions change over time. The PlanetScope analysis uses only the 3 May 2024 assessment as its target because only one suitable pre/post imagery pair is available. The July assessment is used only to identify potential temporal label mismatches caused by the PlanetScope post-event imagery being acquired in early June.

UNOSAT provides geolocated assessment observations classified into five categories: No visible damage, Possibly damaged, Moderately damaged, Severely damaged, and Destroyed. These observations are linked to building geometries from the Microsoft Global Building Footprints dataset. We define Moderately damaged, Severely damaged, and Destroyed as positive damage categories. No visible damage and Possibly damaged are treated as part of the intact class in our binary target. More generally, any retained footprint that is not matched to a positive UNOSAT damage observation remains labelled intact. Consequently, “intact” is an operational category meaning that the footprint was not matched to evidence of moderate-or-greater damage, rather than necessarily an explicit confirmation that the building was undamaged. Treating Possibly damaged as non-damaged is a conservative choice motivated by the lower resolution of our model inputs: if damage cannot be identified confidently using UNOSAT’s sub-metre imagery, it is unlikely to be distinguishable reliably in 3-metre PlanetScope or 10-metre Sentinel-1 imagery.

Satellite imagery We use Sentinel-1 SAR and PlanetScope optical imagery because they contribute different advantages to the study. Sentinel-1 provides the temporal coverage needed to evaluate the aforementioned generalisation across assessment dates. Its freely available and regularly acquired imagery allows us to obtain observations corresponding to the different UNOSAT assessment periods without relying on a single acquisition window. We use Sentinel-1 rather than Sentinel-2 because radar backscatter penetrates cloud cover and is unaffected by illumination, guaranteeing consistent acquisitions over conflict zones where optical imagery is frequently obscured, and because a building’s collapse produces a detectable change in its radar scattering mechanism, transitioning from the double-bounce return of a standing structure to the diffuse return of rubble (Ballinger 2025).

At approximately 10 metres per pixel, however, its spatial resolution is relatively coarse for building-level deep learning. Individual buildings may occupy only a few pixels, making finer damage characteristics such as partial collapse or rubble patterns difficult to resolve. PlanetScope addresses this spatial-resolution limitation. Its approximately 3-metre resolution, together with RGB and NIR bands, provides substantially more detail from which a supervised model can learn visual features of damage that may be lost in Sentinel-1. PlanetScope therefore allows us to investigate whether higher-resolution optical imagery provides information beyond the Sentinel-1 resolution.

The two data sources consequently involve a deliberate trade-off. Sentinel-1 provides the temporal coverage required to assess generalisation across multiple dates, but at lower spatial resolution. PlanetScope provides substantially higher spatial resolution, but our access does not provide the same temporal coverage. In particular, the PlanetScope imagery available for the principal training period was acquired on 1–2 June 2024, approximately one month after the 3 May UNOSAT assessment. We therefore cannot use PlanetScope to reproduce the same sequence of temporally aligned assessments available with Sentinel-1. Furthermore, this temporal mismatch creates a potential source of systematic label noise. A building that was intact on 3 May and therefore labeled as undamaged may have been damaged or destroyed before PlanetScope was acquired in June. The model would then receive an intact label for a structure that appears damaged in the input imagery. The consequences of this mismatch, and the strategy we use to reduce its impact on the training data, are discussed in the pre-processing section.

2.1 Preprocessing

The Sentinel-1 and PlanetScope data are processed separately to account for their different spatial and temporal characteristics. In both cases, building footprints are converted into fixed-size image patches centred on the building centroid.

Building footprints and damage labels. Building geometries are obtained from the Microsoft Global Building Footprints dataset. We remove footprints with an area of 50 m² or less, which are more likely to represent small structures or geometries that cannot be meaningfully resolved in the satellite imagery. UNOSAT damage assessments are provided as geolocated observations that do not always fall exactly within the corresponding building polygon. To accommodate positional discrepancies between the two sources, we place a 10 m buffer around each UNOSAT observation classified as Moderately damaged, Severely damaged, or Destroyed. Any retained building footprint intersecting such a buffer is labelled as damaged. Footprints not matched to one of these positive damage observations—including observations classified as Possibly damaged—remain in the intact class.

Sentinel-1 For each study area, we select a single relative orbit to ensure consistent viewing geometry across observations. The pre-conflict image is constructed as a median composite of acquisitions from the preceding 12 months, while post-conflict imagery uses the acquisition closest to each UNOSAT assessment date within a ± 12 -day window. Both VV and VH polarisations are retained, resulting in four input channels: pre- and post-event VV and VH. For each building, a 32×32 pixel patch is extracted, corresponding to 320×320 m at 10-metre resolution. Channel-wise normalisation parameters are calculated from the training data and applied unchanged to validation and test sets. Additional Sentinel-1 observations at ± 12 and ± 24 days from each assessment date are generated for the stacking model, allowing temporal information to be incorporated without directly training the CNN on these observations. The exact dates and their offsets are provided in Table A.1.

PlanetScope PlanetScope uses four optical bands (blue, green, red, and NIR) from the pre- and post-event imagery, producing eight input channels. Because the imagery used for training was acquired approximately one month after the 3 May 2024 UNOSAT assessment, we use the July assessment to identify potential temporal false negatives in the training data. Specifically, we remove buildings that were labelled intact in May but damaged in July, since these buildings may have been damaged between the May assessment and the June PlanetScope acquisition. Buildings already classified as damaged in May are retained, regardless of their July status. This filtering is intended to remove observations for which the May label is inconsistent with the condition visible in the PlanetScope imagery, while preserving as much of the original training sample as possible. For the remaining buildings, 32×32 pixel patches are extracted around the centroid, corresponding to 96×96 m at 3-metre resolution. Invalid or masked patches are removed. PlanetScope bands are independently normalised using clipping bounds estimated from the training set, which are then fixed for validation and test data.

2.2 Exploratory Data Analysis

The exploratory analysis addresses four questions that inform the subsequent modelling design. First, we examine the spatial and temporal evolution of the UNOSAT damage labels to identify potential inconsistencies between assessment dates and satellite observations. Second, we assess whether the satellite imagery contains systematic signals associated with building damage and whether these signals differ between PlanetScope and Sentinel-1. Third, we quantify the spatial dependence between neighbouring buildings to determine an appropriate data-splitting strategy and assess the risk of spatial leakage. Finally, we examine how the resulting geographic split affects label prevalence and feature distributions, and whether the underlying damage signal remains consistent across the resulting subsets.

Spatial Distribution of Damage and Temporal Mismatch Figure 1 shows the spatial distribution of UNOSAT damage labels across the three assessment dates used in our analysis: 3 May, 6 July, and 6 September 2024. The maps show both the persistence of previously damaged areas and the expansion of damage over time, with the overall share of buildings classified as damaged increasing from 49.8 percent in May to 57.2 percent in July and 59.8 percent in September. The increase between May and July is particularly relevant for the PlanetScope data: because the PlanetScope imagery used for training was acquired in early June, buildings that were intact in the May assessment but classified as damaged by July could already have been damaged in the June imagery while still carrying an intact May label. The spatial distribution also changes over time, with substantial increases in damage in southern areas such as Khan Younis and Rafah. These temporal and spatial shifts motivate both our treatment of potential false-negative labels and the geographic split used to evaluate model generalisation.

Spectral and Radar Signatures of Building Damage To assess whether the satellite imagery contains meaningful signals of building damage, we compare the pre- and post-conflict distributions across the UNOSAT damage categories. The underlying band-level distributions are shown in Figure A.1, while Figure A.2 and Figure A.3 summarise the pre-post differences in the distributions for each category using the Wasserstein distance, NDVI change, and mean optical and SAR signal strength. Together, these measures assess whether the imagery captures systematic changes associated with damage severity and if the footprints that got newly damaged in July already contain signals in the PlanetScope data.

The results show a clear severity gradient in the PlanetScope data. The Wasserstein distance generally increases from *No visible damage* to *Destroyed*, with the strongest separation in the red and NIR bands. Notably for the PlanetScope data, the largest increase in separation typically occurs between *Possibly damaged* and *Moderately damaged*. This supports our aggregation of the UNOSAT categories into two classes for the PlanetScope models, with *No visible damage* and *Possibly damaged* grouped as *intact*, and *Moderately damaged*, *Severely damaged*, and *Destroyed* grouped as *damaged*. Sentinel-1 shows a different pattern: VV and VH exhibit limited separation among *No visible damage*, *Possibly damaged*, and *Moderately damaged*, but substantially stronger separation for *Severely damaged* and *Destroyed*. This is consistent with optical imagery capturing more gradual changes in building and surface appearance, while SAR is primarily sensitive to more substantial structural changes.

Importantly, the grouped category of buildings classified as *No visible damage* in May but subsequently identified as damaged in July also exhibits a clear signal in the June PlanetScope imagery. Its optical distributions are shifted relative to the *No visible damage* category, with particularly pronounced differences in the red and NIR bands. This provides empirical evidence that some buildings labelled as intact in May had already undergone detectable changes by the time the June imagery was acquired, supporting the temporal-mismatch concern identified above. In contrast, the same July-newly-damaged group shows little separation from the intact categories in Sentinel-1. This is consistent with the absence of a temporal mismatch for the Sentinel-1 data.

Overall, both modalities contain meaningful damage-related information, but capture different aspects of destruction. PlanetScope provides a stronger and more gradual separation across damage severity and also captures changes associated with damage occurring between the May and July assessments. Sentinel-1 provides a more pronounced signal for severe structural destruction while retaining the temporal alignment needed to evaluate changes across the different UNOSAT assessment dates.

Furthermore, in the Appendix, Figure A.4 provides representative before–after PlanetScope patch pairs. Even at 3-metre resolution, some changes associated with destruction are visually apparent, although the coarse spatial resolution makes finer forms of damage difficult to identify reliably by visual inspection alone.

Spatial dependence and split design

2.2.1 Spatial Dependence and Split Design

A geographic split is necessary to ensure that model performance reflects transferable damage signals rather than similarity between neighbouring buildings. Damage is strongly spatially autocorrelated: across 228,725 footprints, the damage status of a building correlates by 0.808 with the average label of its 20 nearest neighbours ($k = 20$). A random split would therefore place many highly similar buildings in both the training and test sets, creating the potential for the model to exploit local spatial context rather than generalise to a new geographic region.

We therefore divide the study area into four latitude-ordered bands: training, stacking, validation, and test, with the northern region used for model development and the southern region held out for the final evaluation. The same geographic split is used across all three architectures: a SAR-only model, a multispectral PlanetScope model, and a PlanetScope RGB model with a pretrained backbone (see Figure B.1, 2(a), and 2(b)). Stacking is used for the from-scratch SAR and PlanetScope models, while the pretrained RGB model is trained without stacking.

The geographic separation is particularly important because each observation is represented by a spatial patch rather than by the building footprint alone. PlanetScope inputs use 32×32 pixel patches corresponding to 96 m per side, while Sentinel-1 inputs cover 320 m per side. Two observations can therefore contain overlapping image information even when their building footprints are distinct. We consequently define potential spatial leakage based on patch overlap rather than on an arbitrary distance threshold.

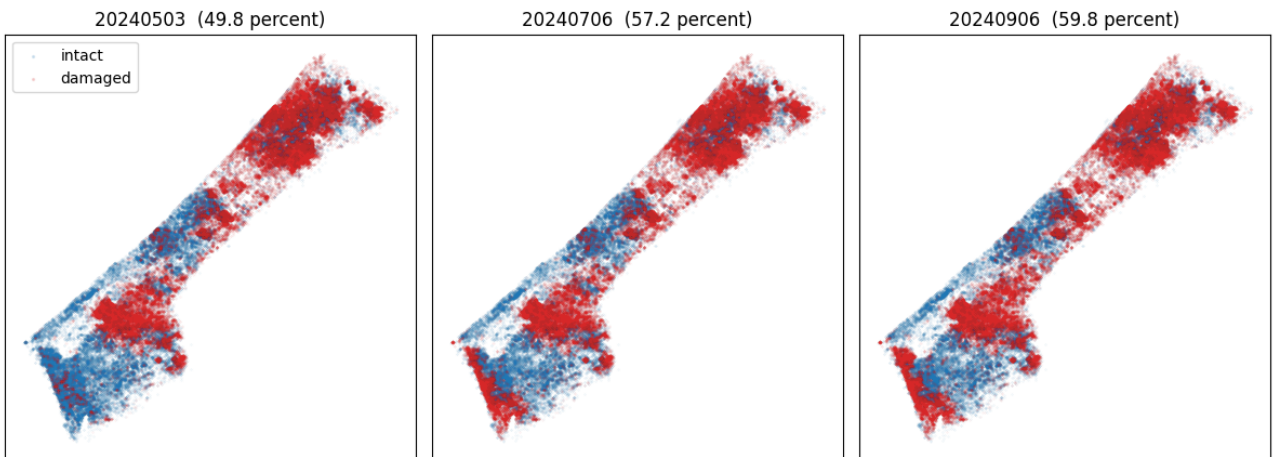


Figure 1: UNOSAT damage labels for 2024-05-03, 2024-07-06, and 2024-09-06.

The difference between random and geographic splitting is substantial. Under a random split, the median distance between footprints is only 16.3 m, and 99.2% of test footprints overlap with a training footprint at the PlanetScope patch scale, while 100.0% overlap at the Sentinel-1 patch scale. In contrast, the geographic train-test boundary contains an 8,760 m gap, equivalent to 27 times the Sentinel-1 patch width, resulting in 0.0% patch overlap for both modalities. The final train-to-test evaluation is therefore spatially separated at both input resolutions.

The intermediate subsets are not fully spatially independent, however. The train-stack, stack-validation, and validation-test boundaries are adjacent, with 0–1 m separation. Patch overlap across these boundaries is 2.8%/10.2%, 0.7%/2.8%, and 1.3%/4.8% for PlanetScope/Sentinel-1, respectively. Thus, while the geographic split provides a leakage-free final train-to-test evaluation, the internal boundaries retain limited overlap. A margin of at least 320 m between adjacent subsets would be required to eliminate patch overlap entirely.

Implications of the Geographic Split The geographic split has important implications for model evaluation. Because the four sets correspond to geographically distinct regions rather than random samples of the same spatial distribution, we expect differences in both label prevalence and input-feature distributions. Figure A.6 and Figure A.5 illustrate these differences and assess whether the underlying relationship between the imagery and building damage remains consistent across splits.

Damage prevalence varies substantially across the four sets, ranging from approximately 24% in the stacking set to 66% in the training set, compared with 53.8% across the full dataset. The distributions of the spectral and SAR features also differ between splits, reflecting differences in geographic context, land cover, and damage history. These shifts are an expected consequence of the geographic split and make the held-out test set a more demanding evaluation of geographic generalisation than a random split would provide.

Despite these distributional shifts, the damage-related signal remains visible across all splits. In particular, Figure A.6 shows that damaged buildings consistently exhibit a more negative NDVI change than intact buildings in the training, stacking, validation, and test sets. Thus, while the geographic split changes the overall distribution of both labels and features, the relationship between the imagery and damage persists in the held-out region. This provides a basis for evaluating whether the models can recover a transferable damage signal rather than relying on region-specific distributions.

3 Results

3.1 Models and Training

From-Scratch Models Our from-scratch models can be distinguished by architecture and data. We train from-scratch models using a conventional CNN structure (Base CNN) as well as a siamese approach (Siamese CNN). The Base CNN contains two layers at eight filters each, each a 3×3 convolution followed by batch normalization, ReLU activation, and max pooling. Its head performs global average pooling, flattens the output, applies dropout, and passes the result through a linear layer producing a single logit. At 1,529 (SAR) and 1,817 (Planet) parameters¹, it serves as the low-complexity baseline. The Siamese CNN uses a shared-weight encoder, structured identically to the Base CNN’s convolutional stack, applied separately to the pre-event and post-event channels. Its head applies dropout to the concatenation of the resulting pre- and post-event outputs, and their signed difference, then passes this through a linear layer, a ReLU activation, dropout again, and a final linear layer producing one logit. This gives the classifier explicit access to the individual pre- and post-event representations as well as the change between them, rather than requiring the network to infer that change implicitly, as Base CNN must. It has 7,769 parameters for SAR and 7,913 for Planet data. Both architectures are trained with class-weighted binary cross-entropy, the positive class weighted by the ratio of undamaged to damaged buildings in the training band.

During HPO, we additionally optimize model architecture – number of filters, depth, and dropout (plus head dimension, and for Planet’s siamese head, head dropout) – alongside the optimizer hyperparameters; see Table B.1. Further, we employ a stacking variant where an XGBoost model uses inputs from the corresponding CNN to infer gains from spatial and temporal smoothing. It is provided with the confidence score of the building to be predicted, the mean and standard deviation of the 8 and 32 nearest buildings, confidence scores of 4 offsets (12-day difference each, following the S1 orbit), the change compared to the average of the two previous offsets, and the change compared to the average of the two next offsets. This second-stage model is likewise class-weighted, via `scale_pos_weight` computed on the stacking band. Since our Planet data offers only a single pre/post pair rather than repeated assessment dates, temporal smoothing is only available for SAR.

Splitting As previously motivated, we geographically split Gaza into three/four bands: train, val, stack, and test, see Figs. B.1 and 2(a). Given our preprocessing steps due to the date differences, damage prevalences differ slightly between the Sentinel and Planet data.

Training The baseline (hand-set) hyperparameters are identical across SAR and Planet: batch size 128, learning rate 1×10^{-3} , weight decay 1×10^{-4} , AdamW optimizer, up to 120 epochs with early stopping after 10 epochs without

¹ SAR data has 4 channels (VV and VH, each for the pre- and post-event image). Planet has 8 (R, G, B, and NIR, each for the pre- and post-event image), so its Base CNN has more parameters.

validation-ROC-AUC improvement. The threshold used for evaluation is calibrated on the validation set using best-F1. Planet training additionally applies geometric augmentation: each pre/post pair receives an identical random horizontal flip, vertical flip, and multiple-of-90° rotation, applied to the stacked pair so the two acquisitions stay in register. SAR training uses no augmentation.

We do not observe the extreme positive-class rarity often encountered in war-damage assessment, although damage prevalence varies substantially across geographic bands and assessment dates (Figures B.1, 2(a), 2(b)). We therefore use ROC AUC as the primary model-selection criterion because it evaluates the ranking of damaged and intact buildings across all possible thresholds and is less directly affected by class prevalence than F1 or PR AUC. F1 depends on a specific decision threshold and is used only after model selection to calibrate that threshold on the validation data.

As a secondary check specific to the Sentinel-1 experiments, we evaluate cross-context generalisation in four held-out cities: Raqqa in Syria, Mosul in Iraq, and Chernihiv and Rubizhne in Ukraine. These cases differ from Gaza in building characteristics, climate, conflict conditions, and damage prevalence, which ranges from 2.7% to 41%. Because the models were developed and selected using Gaza data, this analysis constitutes a demanding transfer test rather than an evaluation setting for which strong performance was expected. We anticipate the greatest degradation in Chernihiv, where damaged buildings are particularly rare.

For hyperparameter optimization we use Optuna’s TPESampler for 10 trials, each training the CNN on a 50% subsample of the training band for up to 25 epochs, with that trial’s own training stopped early after 6 epochs without validation-ROC-AUC improvement – the same early-stopping mechanism as the full run, just on a smaller time budget to keep training times manageable. Beyond the architecture parameters listed above, HPO also searches learning rate, weight decay, and batch size. The exact optimization intervals are given in Table B.1. Evaluation and selection are based on validation ROC-AUC throughout.

The XGBoost stacker is trained on the stacking band – rather than the CNN’s own training band – to avoid learning from over-confident CNN scores it was never asked to generalize beyond. Its early stopping likewise uses the validation band, and it too is optimized toward ROC-AUC. Additionally, we implement a fixed neighbor rule which takes the eight nearest neighbors and weights their scores 30%-70% to that of the building to be predicted. It serves a baseline check of whether the additional complexity of an XGB model is justified.

We evaluate all configurations produced by this pipeline: the raw CNN score, the fixed-weight spatial-neighbor control, and the learned XGBoost stacker with spatial (and, for SAR, spatial+temporal) features, each under both the baseline and HPO-tuned CNN. We choose the final best configuration based on validation ROC-AUC and evaluate it once on the test set.

Planet Pre-Trained To compare against our from-scratch models, we fine-tune a pretrained encoder that has been trained on general satellite imagery. For this we use a ResNet-50 initialised from the Remote Sensing Pretraining (RSP) checkpoint of Wang et al. (2023) (the 300-epoch MillionAID model, taken at its final epoch), which was pretrained on the MillionAID scene-classification dataset comprising satellite imagery of varying type and resolution, including a 3 m resolution similar to the Planet data we are working with. The checkpoint is RGB, so this experiment uses only the red, green and blue Planet bands, while the closest comparable from-scratch model uses NIR imagery additionally. We weighed using pretrained models that included NIR imagery in their training data, but decided a more generalised, larger pretrained model on RGB should outdo them.

Splitting In comparison to the from-scratch CNN, the pretrained model does not require a stacking partition, so the stacking band of that experiment is free to be reused. To keep the results comparable we adopt the identical geographic partition and merge the stacking band into validation, see Fig. 2(b). Normalisation statistics and the class weight are derived from the training band alone.

Model The classifier is a siamese network in which a single shared encoder is applied to both the pre-event and the post-event RGB patch. We use the pretrained encoder, but discard the MillionAID classification head and replace the final fully connected layer by an identity mapping, so that the encoder outputs a 2048-dimensional pooled feature vector. Reflectances are first rescaled and clipped to the unit interval using one lower/upper percentile pair per band — the 2nd and 98th percentiles, fitted over pre- and post-event pixels jointly from 20,000 sampled training patches — so that both acquisitions share a single radiometric mapping. Within the model, each 32×32 px patch, covering 96 m at the 3 m Planet resolution, is bilinearly upsampled to 128×128 and standardised with the channel-wise mean and standard deviation used during pretraining; the upsampling size is a free parameter, and 224 px would reproduce the pretraining resolution exactly at roughly three times the computational cost. The pre- and post-event patches are concatenated along the batch dimension, encoded in a single forward pass and separated afterwards, which is possible because the batch-normalisation layers operate in inference mode throughout and therefore cannot mix statistics between the two images. The resulting feature vectors are combined as their concatenation together with their absolute difference, yielding a 6144-dimensional representation that is passed to a classification head consisting of dropout (0.4), a 256-unit fully connected layer with ReLU activation, dropout (0.3), and a single-unit output producing one logit per building.

Training Fine-tuning proceeds in two phases. In the first, the entire backbone is frozen and only the classification head is trained, for up to 20 epochs at a learning rate of 1×10^{-3} . In the second, the phase-one optimum is used as the starting point and the third and fourth residual stages of the backbone — approximately 94% of its parameters — are unfrozen and trained for up to 30 epochs at a reduced learning rate of 1×10^{-5} , while the stem and the first two residual stages remain fixed. Batch-normalisation parameters and running statistics are frozen in both phases, since the

Table 1: Validation ROC AUC and F1 by pipeline stage on the Sentinel-1 SAR dataset (mean across the three validation assessment dates).

Pipeline stage	Metric	Base CNN		Siamese CNN	
		Baseline	HPO	Baseline	HPO
CNN score only	ROC AUC	0.8528	0.8668	0.8579	0.8581
	F1	0.8545	0.8715	0.8673	0.8665
+ fixed neighbor rule	ROC AUC	0.8553	0.8687	0.8594	0.8598
	F1	0.8565	0.8732	0.8682	0.8682
+ XGBoost (spatial)	ROC AUC	0.8620	0.8740	0.8613	0.8613
	F1	0.8649	0.8793	0.8725	0.8721
+ XGBoost (spatial + temporal)	ROC AUC	0.8688	0.8790	0.8649	0.8683
	F1	0.8720	0.8832	0.8761	0.8774

Table 2: Final sealed test-set performance on the Sentinel-1 SAR dataset. Base CNN is thresholded at 0.548; Siamese CNN at 0.407 (both chosen on validation, pooled best-F1). $F1_{\text{oracle}}$ is the best-possible F1 at the optimal threshold for that date (computed using test labels). Dates are for 2024.

Model	Date	Prevalence	ROC AUC	PR AUC	Precision	Recall	F1	$F1_{\text{oracle}}$
Base	05-03	30.9%	0.893	0.839	0.808	0.742	0.773	0.774
	07-06	48.0%	0.893	0.889	0.833	0.797	0.814	0.819
	09-06	52.2%	0.899	0.907	0.836	0.840	0.838	0.840
Siamese	05-03	30.9%	0.881	0.821	0.622	0.827	0.710	0.753
	07-06	48.0%	0.888	0.882	0.765	0.862	0.811	0.812
	09-06	52.2%	0.893	0.903	0.789	0.875	0.830	0.833

batch size is too small to re-estimate them reliably. Both phases use the Adam optimiser, a batch size of 64, and a binary cross-entropy objective in which the positive class is weighted by the ratio of undamaged to damaged buildings in the training band; because that band is damage-majority, the resulting weight is 0.478 and thus down-weights the damaged class. Mixed-precision arithmetic with gradient scaling is used, with the loss itself accumulated in full precision. Validation ROC-AUC is the monitored quantity and governs early stopping (patience of 8 epochs, with the best weights restored), learning-rate reduction on plateau (factor 0.5, patience of 4 epochs, lower bound 1×10^{-7}), and checkpointing of the best epoch. During training, each pre/post pair receives an identical random horizontal flip, vertical flip and multiple-of-90° rotation, each transform being applied to the stacked pair so that the two acquisitions stay in register; no photometric augmentation is applied.

3.2 Evaluation

SAR Table 1 shows validation results. ROC AUC and F1 move together across every pipeline stage for both architectures, so more complex configurations do not trade one metric for the other. More complex configurations consistently improve both metrics for the Base CNN. For the Siamese CNN, additional pipeline complexity also improves both metrics within a given run. HPO tuning itself, however, does not consistently help at a fixed pipeline stage. ROC AUC under HPO is never worse than the baseline CNN. F1, however, is lower under HPO at the raw-CNN and spatial-stacking stages, and only improves once temporal stacking is added.

The improvement for the Base CNN is itself marginal relative to its cost. The raw HPO-tuned Base CNN increases ROC AUC by 0.014 and F1 by 0.017, for a 13-fold increase in parameters (1,529 to 19,937, see Table C.1). For the Siamese CNN the gains are smaller and absent entirely in one configuration, while its parameter increase is larger (15-fold, to 118,113) and disproportionate to the benefit. The learned spatial stacker clearly earns its place over the fixed-weight neighbor rule, outperforming it at every stage and improving further given temporal information. As previously explained, the latter comparison is only possible for SAR. Planet has a single pre/post image pair, so temporal-offset features do not apply. Additionally, its validation ladder skips the fixed-neighbor-rule step, going straight from the raw CNN score to the learned spatial stacker. Since the fixed rule is essentially a hand-set version of what the stacker’s neighbor features can already learn, the SAR result suggests Planet loses little by skipping it, beyond a simple baseline for comparison.

For our final test evaluation, we choose the configuration with the best validation performance: the HPO-tuned CNN with the full spatial-temporal XGBoost stacker. Results can be found in Table 2. Precision, recall, F1, and PR-AUC all rise steadily with prevalence across the three dates, while ROC AUC stays nearly flat. As Fig. 1 shows, damage in southern Gaza increased substantially over the recorded period, driving this rising prevalence. With the exception of recall, Base outperforms the Siamese configuration on every metric. May exposes the biggest weakness of the Siamese network: Precision degrades sharply at a lower prevalence to just 0.622 compared to 0.808 for the Base CNN. The May assessment is the clear outlier on every threshold-dependent metric. The Base CNN’s confusion matrices (Fig. C.1) show a relative false-negative bias that shrinks over time: 46% more false negatives than false positives in May, 27% more in July, and roughly even by September. The theoretically optimal (oracle) threshold does not close this gap either: $F1$ and $F1_{\text{oracle}}$ differ by at most 0.005 across all three dates, meaning the frozen decision threshold is not the cause. We instead hypothesize this is because the CNN and the XGBoost stacker rely on similar spatial-clustering information – 320 m patch include neighboring buildings similar to the k-nearest-neighbors used for XGBoost. This information is less useful when damage is more spatially scattered, as it plausibly was earlier in the conflict, likely driving this performance decrease. In contrast, the Siamese CNN benefits noticeably from an optimized threshold for the May date (0.710 vs 0.753). Despite similar PR-AUC values to the Base CNN, it shows a strong FP-bias, see Fig. C.2.

Table C.3 extends this evaluation to four unseen cities. The Base CNN pipeline generalizes best to Raqqa and Rubizhne, where prevalence (41% and 25%) is closer to Gaza’s training range, though F1 is much lower here at 0.654 and 0.492, respectively. Performance particularly degrades on the two lowest-prevalence cities, most severely on Chernihiv (2.7% damaged). There, ROC AUC remains informative (0.754), but F1 collapses (0.093, oracle 0.250) – the same ROC-AUC/F1 divergence discussed above, now extreme enough that even the best possible threshold cannot recover a usable operating point.

Planet Validation results can be found in Table 3. The larger of the two gains comes from tuning the CNN itself rather than from the spatial stacker that follows it. The raw HPO-tuned Base CNN improves on the hand-set baseline by 0.051 ROC AUC and 0.038 F1, at the cost of a 54-fold increase in parameters (1,817 to 98,769, see Table C.4). Unlike the SAR pipeline, where a comparable increase in complexity hit an overfitting ceiling earlier, additional capacity is clearly rewarded here, which we attribute to Planet’s finer spatial resolution offering the network more structure to exploit. Fig. C.6 illustrates the same asymmetry: the two HPO curves are nearly indistinguishable from one another and sit well above both baseline curves, indicating that hyperparameter tuning, not the second-stage model, is what separates strong from weak configurations on this dataset.

Table 3: Validation ROC AUC and F1 by pipeline stage on the Planet dataset (single-date, spatial-only stacking; no temporal offsets are available). “Baseline” uses hand-set CNN hyperparameters; “HPO” uses the Optuna-tuned CNN.

Pipeline stage	Metric	Base CNN		Siamese CNN	
		Baseline	HPO	Baseline	HPO
CNN score only	ROC AUC	0.8751	0.9262	0.8917	0.9208
	F1	0.8602	0.8979	0.8769	0.8856
+ XGBoost (spatial)	ROC AUC	0.8836	0.9300	0.9006	0.9306
	F1	0.8693	0.9020	0.8864	0.9016

We select the same class of configuration used for SAR, the HPO-tuned CNN with the spatial stacker, and evaluate it once on the sealed test set, see Table 4. Notably, this configuration is essentially indistinguishable from the Siamese CNN on validation ROC AUC (0.930 versus 0.931), yet the two architectures separate clearly on test (0.935 versus 0.912). This divergence between validation parity and test superiority echoes the pattern already noted for SAR, and reinforces that validation agreement between two candidate architectures does not guarantee equivalent generalization once evaluated on unseen ground. Like for our SAR configurations, we observe a pattern where the Siamese CNN outperforms the Base CNN on recall, but significantly loses out on precision (0.786 vs 0.694). It shows the same FP-bias and, given its high PR AUC, would profit from a different threshold, see Fig. C.8. Both architectures select their final decision threshold the same way, by maximizing F1 on pooled validation predictions of the same final ensemble (CNN plus spatial, and for SAR spatial-temporal, XGBoost stacking). Despite this identical procedure, the resulting thresholds land in noticeably different places relative to each model’s own intact-class scores, as Figures C.9 and C.10 show.

On SAR, at the earliest test date, Base’s threshold (0.548) sits well above its mean intact score (0.30), while Siamese’s threshold (0.407) sits much closer to its own mean intact score (0.33). The same pattern appears on Planet: Base’s threshold (0.613) again sits comfortably above its mean intact score (0.32), while Siamese’s threshold (0.480) sits only slightly above its own mean intact score (0.35). In both figures, the dashed threshold line cuts through a visibly larger share of the blue intact distribution in the Siamese row than in the Base row, leading to a larger share of false positives we observe for Siamese across both datasets.

Table 4 places this result alongside the pretrained RSP ResNet-50, which, despite roughly 250 times as many parameters as the Base CNN, does not surpass it on any metric, suggesting that task-specific architecture can outweigh

Table 4: Final test-set results, Planet dataset (Prevalence=37.3%).

Model	ROC AUC	PR AUC	Precision	Recall	F1
Base CNN	0.935	0.909	0.786	0.871	0.826
Siamese CNN	0.912	0.881	0.694	0.884	0.777
RSP ResNet-50	0.905	0.879	0.736	0.836	0.783

generic pretraining for this classification problem. Fig. 2 shows the Base CNN’s final test confusion matrix and precision-recall curve. At the frozen threshold of 0.613, the model errs more often toward false positives (4,940) than false negatives (2,681), indicating a mild preference for flagging uncertain buildings as damaged rather than overlooking them. An example visualization of the Base CNN predictions overlaid on Sentinel 2 imagery, can be found in Fig. C.11.

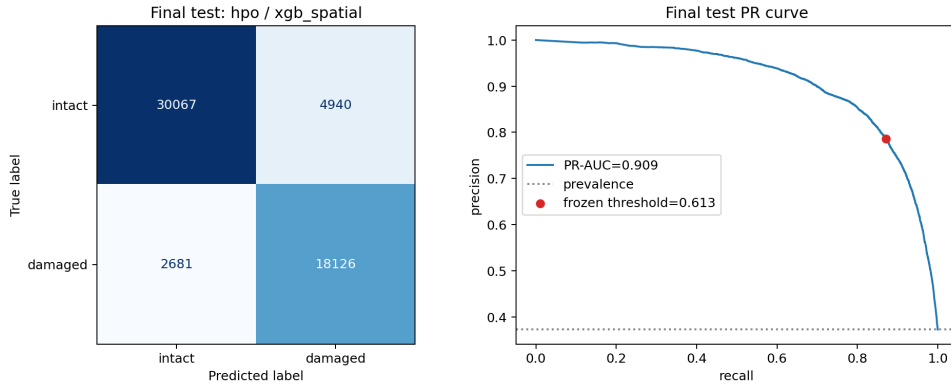


Figure 2: Final test confusion matrix and precision-recall curve for the Base CNN’s selected pipeline (HPO CNN with spatial XGBoost stacker), at the frozen threshold of 0.613.

4 Damage to Health Facilities in Gaza

The pipeline evaluated above assigns a damage score to every building footprint in Gaza. This section demonstrates one substantive use of that output, and returns to the motivation given in the introduction: monitoring damage to specific categories of buildings rather than only the aggregate. We place health facilities within the predicted damage surface and ask whether their damage differs from that of the buildings immediately around them. This is an example application of the classifier, not a further evaluation of it. Model performance was established in Section 3.2 on the validation and test bands and is taken as given.

Facility data Facility locations are taken from the [OCHA \(2023\)](#) health facility dataset for Gaza, harmonised into a single layer of 73 primary care sites and 28 hospitals, 101 facilities in total. OCHA records each facility as a point rather than a building outline, so a facility must first be linked to a footprint before it can inherit a damage call. Because the correct linkage is not observable, we use two rules and report both.

Under the *nearest rule*, the facility takes the score of the single footprint closest to its recorded point and is classified as damaged if that score exceeds the frozen decision threshold of 0.20 calibrated for the CNN-only architecture in Section 3.2. Under the *frame rule*, every footprint within 100 m of the point is scored and thresholded individually, and the facility is classified as damaged if at least 70 percent of those footprints are. The nearest rule is the more faithful reading of the individual facility, but rests on a single footprint that the OCHA point may not in fact identify. Where the two rules agree, the resulting call does not depend on which footprint was selected; in the event they agree on 95 percent of facility-date observations. Both are applied at each of the three assessment dates.

Results Figure 3 shows each facility’s call on top of the footprint-level predictions. The picture varies sharply along the strip: in the north nearly every facility is called damaged, the central corridor is mixed with a visible cluster of intact calls, and the south is mixed again. Part of this gradient is an artefact of the geographic split rather than of the war: the training band covers northern Gaza, so the model has already seen the footprints it is most confident about, and calls there should be treated as optimistic. Rather than discarding these facilities, we report all 101 and carry the split band through every table.

Under the nearest rule, the share of facilities classified as damaged rises from 61.4 percent on 3 May to 64.4 percent in July and 77.2 percent in September; 18 facilities move from intact to damaged over the period, and 3 show the reverse transition, which, since dates are scored independently, is classifier noise rather than recovery. Within the held-out test band alone the share rises from 25.0 to 70.8 percent. Facility surroundings are also more damaged than the city at large: averaged over each facility’s 100 m frame, the damaged share exceeds that of a random held-out footprint sample by 18–22 percentage points at every date. Facility blocks are, however, urban and central while the city-wide sample is not, so this excess conflates the facilities with where they stand; the comparison that follows removes location.

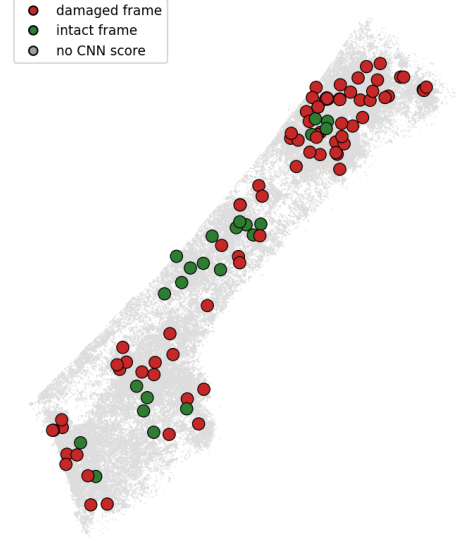


Figure 3: Health facilities overlaid on building-level damage predictions, 6 September 2024

Facilities relative to their surroundings We compare each facility with its own neighbours: footprints within 60 m of the facility point are the cases, footprints out to 500 m the controls, matched on area and assigned to their nearest facility. A Cochran–Mantel–Haenszel odds ratio pools one stratum per facility and date, with a cluster bootstrap over facilities for the interval. Because case and control footprints within a stratum share the same location, they also share the same split band, so band-level optimism affects both arms alike and all bands enter the comparison.

The pooled odds ratio is 0.87, with a 95 percent interval of 0.54 to 1.49, and lies below one at two of the three assessment dates (Table 5). In probability terms the contrast is modest: 63.8 percent of facility footprints are called damaged against 70.3 percent of matched neighbours, a difference of 6.4 percentage points; at damage prevalence this high the odds ratio overstates the gap, so we report both. Facilities are thus classified as damaged somewhat less often than their neighbours, but the difference is not distinguishable from chance. Two biases pull in known directions: classifier error attenuates the odds ratio toward one, and any survivorship in the facility list pushes it below one. We make no claim about intent.

Table 5: Facility footprints (fac.) against size-matched neighbouring footprints (nbr.) at the frozen threshold, CNN calls only, all bands (101 facilities, 300 strata)

Dates	%dmg fac.	%dmg nbr.	pp diff	OR [95% CI]
Pooled	63.8	70.3	−6.4	0.87 [0.54, 1.49]
3 May	53.2	62.8	−9.7	0.73 [0.44, 1.27]
6 July	64.2	71.3	−7.1	0.81 [0.43, 1.47]
6 September	74.1	76.7	−2.6	1.13 [0.61, 2.20]

Further Applications The application shown here is a demonstration rather than a use case in its own right: Gaza is the city the model was trained on, and UNOSAT ground truth exists for every assessment date, so nothing in this section could not have been read off the labels directly. The value of the pipeline lies in settings where no such labels exist, and there the two data sources are not interchangeable. Planet is the stronger classifier, but it is access-restricted, and in our case available only at a single date; Sentinel-1 is free, global, cloud-independent, and revisits every few days. Accepting the accuracy cost documented in Section 3.2 therefore buys something Planet cannot offer: the SAR track can be scored at every acquisition rather than only at the sparse UNOSAT assessments, turning the facility-level calls of this section into a dense time series and allowing damage around specific infrastructure to be monitored close to real time in any conflict setting.

5 Discussion and Conclusion

Across both data types, the simplest model we tested, the Base CNN, matched or exceeded the performance of the more elaborate Siamese architecture and the considerably larger pretrained ResNet-50. Additional complexity yielded modest gains in some configurations, but typically not proportionate to the added parameter cost. For this task, a compact model trained directly on the problem outperformed both a more elaborate architecture and a substantially larger pretrained one.

The principal contribution of this approach lies not in replacing UNOSAT’s assessments but in complementing them. UNOSAT’s damage surveys depend on manual review of very-high-resolution commercial imagery and are released only every two to three months. Our pipeline instead relies on regularly available satellite imagery and, once trained, can score every building footprint in a study area within minutes. This capability supports more granular monitoring between official assessments and enables, for instance, a semi-automated first pass that flags likely damage for subsequent human review, rather than requiring exhaustive manual labeling. The health facility analysis illustrates this application: instead of an aggregate damage count, the same pipeline can address a targeted question concerning a specific category of buildings, such as hospitals or schools, at a potentially lower labeling cost such an inquiry would otherwise require.

Our models generalize well within Gaza, across both geography and time, the setting for which they were designed and tuned. We additionally assessed transfer to four other cities as a smaller, secondary check, and performance there proved considerably weaker, most notably in Chernihiv. This outcome is unsurprising. The model was developed specifically for Gaza rather than for cross-conflict transfer, and a detector trained on one city’s building stock, imagery conditions, and damage patterns should not be expected to generalize directly to cities with markedly different architecture, climate, and damage prevalence.

Limitations A number of limitations warrant explicit acknowledgment. Our damage labels derive from UNOSAT assessments, which we treat as ground truth throughout the analysis. UNOSAT’s own classifications carry inherent uncertainty, which is why we classify “Possibly damaged” as intact, yet we do not validate our binary labels against another source, so errors in the underlying assessments propagate directly into our results. The temporal stacker also requires future offsets of 12 and 24 days, meaning a prediction with this model requires as 24 days latency. A related concern is that the PlanetScope imagery used for training was acquired approximately one month after the UNOSAT assessment to which it is matched. We removed the clearest instances in which a building was recorded as intact in May but damaged by July, though this correction addresses only one direction of the problem and likely leaves some mislabeled buildings in the training data. Our decision thresholds are likewise fitted once on validation data and subsequently held fixed. Even within Gaza, a single threshold performs unevenly across assessment dates, most notably in May, where precision and recall for the Siamese CNN fall well below their later values, indicating that a threshold calibrated at one point in time should not be assumed to hold under distributional change without recalibration. Similarly, we lack a confirmed explanation for why the May assessment proves harder to predict than subsequent ones. We advance a hypothesis concerning the spatial clustering of damage, but have not tested it directly. Finally, our hyperparameter search was modest in scope, comprising ten trials on a subsample of the training data, so the configuration we describe as tuned represents the best outcome of a limited search rather than a thoroughly optimized one, and a more extensive search could plausibly yield superior configurations for either architecture.

This project engages with data documenting real destruction and, by extension, real human suffering. The underlying UNOSAT and facility records describe an active conflict, and any tool built upon them carries a risk of being applied, or misapplied, beyond the scope for which it was developed. We therefore regard this work as a methodological contribution and a proof of concept for assisted, more frequent damage monitoring rather than as a ready-made operational instrument.

Our cross-city evaluation indicates that the models developed for Gaza do not transfer reliably to other conflict settings. Performance deteriorated substantially in the four held-out cities, particularly where building characteristics, imagery conditions, and damage prevalence differed from those observed in Gaza. The models presented here should consequently be treated as region-specific. Applying the approach elsewhere would require training or fine-tuning with locally relevant imagery and damage labels, followed by context-specific validation and threshold calibration. Without these steps, the predictions should not be used to inform operational or policy decisions.

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Online Appendix for ‘Quantifying War Damage from Satellite Imagery’

Nils Neuner¹ Marcel Kalb² Mike Barth³ Sam Känner⁴

¹ nils.neuner@uni-konstanz.de
² marcel.julian.kalb@uni-konstanz.de
³ mike.barth@uni-konstanz.de
⁴ sam.kaenner@uni-konstanz.de

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A EDA

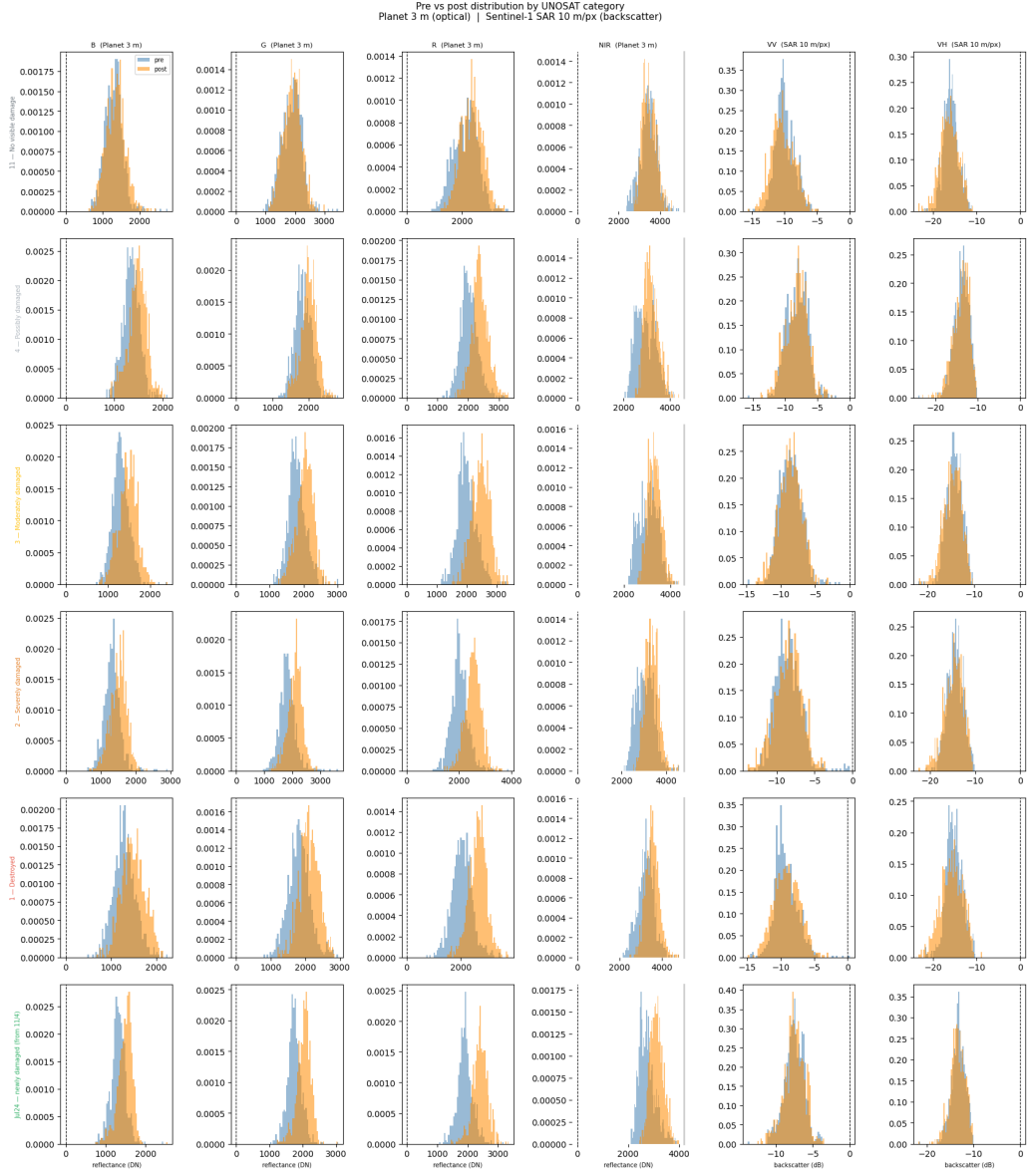


Figure A.1: Pre- vs. post-war distribution of mean patch reflectance/backscatter by UNOSAT damage category (rows) and band (columns), PlanetScope B/G/R/NIR (left four columns) and Sentinel-1 SAR VV/VH (right two columns).

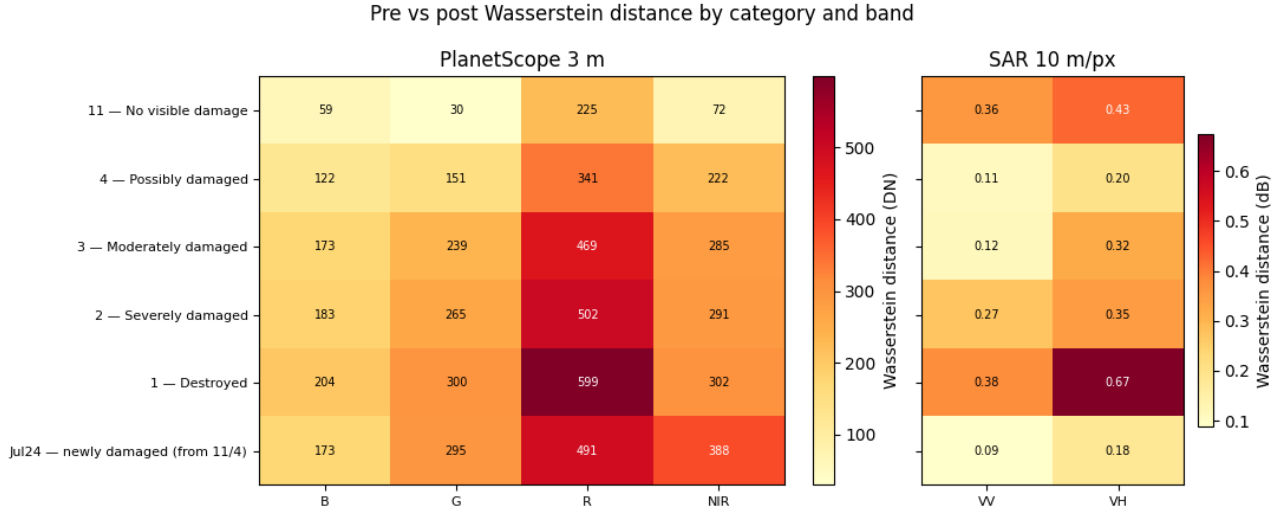


Figure A.2: Pre-vs-post Wasserstein distance by UNOSAT damage category and band, PlanetScope (left, DN) and Sentinel-1 SAR (right, dB). Distance increases monotonically with damage severity for both optical and SAR channels.

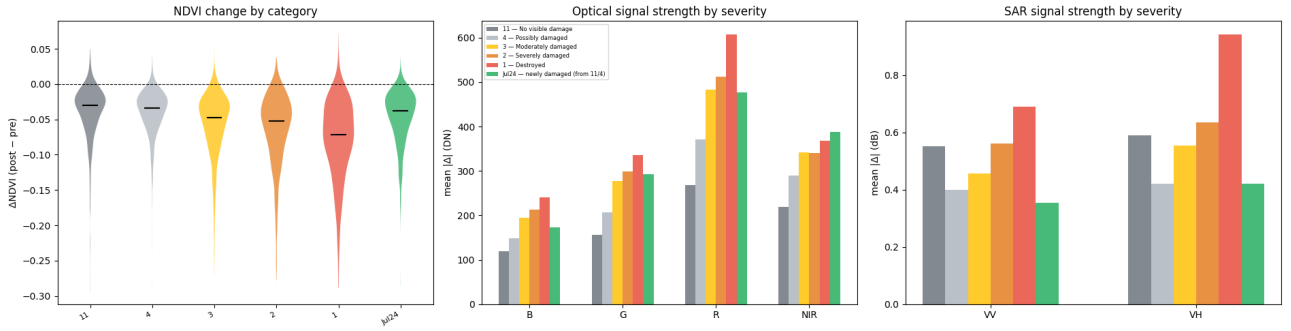


Figure A.3: NDVI change by damage category (left), mean absolute optical band change by severity (middle), and mean absolute SAR band change by severity (right).

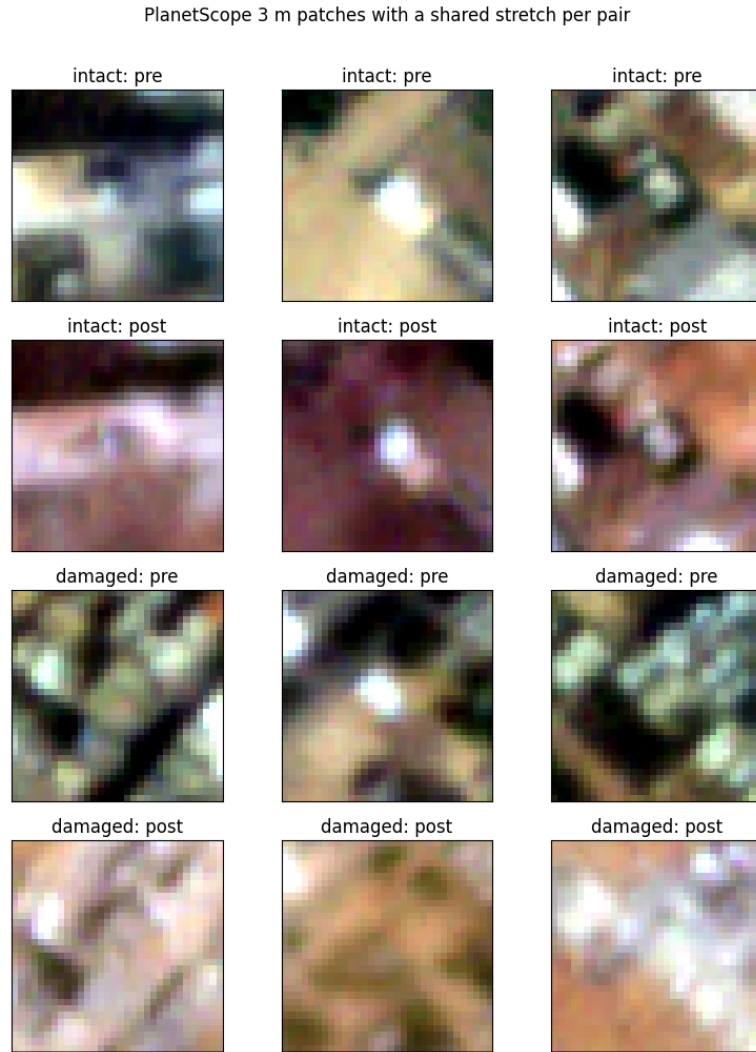


Figure A.4: Example PlanetScope 3 m pre/post patch pairs for intact (top) and damaged (bottom) footprints, with a percentile stretch shared within each pre/post pair.

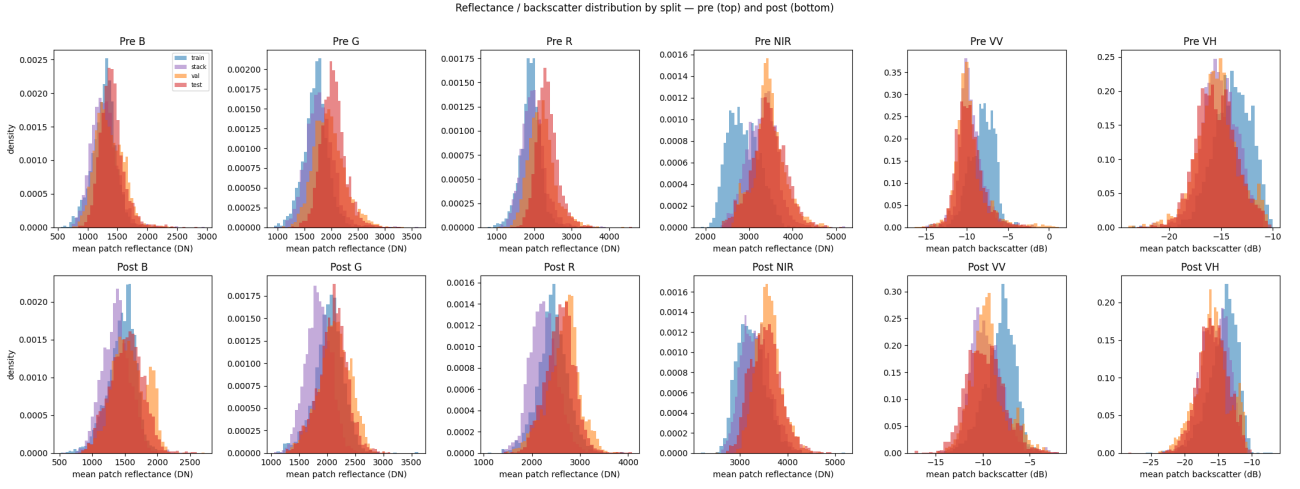


Figure A.5: Pre-war (top) and post-war (bottom) mean patch reflectance/backscatter distributions by band, split by train/stack/val/test assignment.

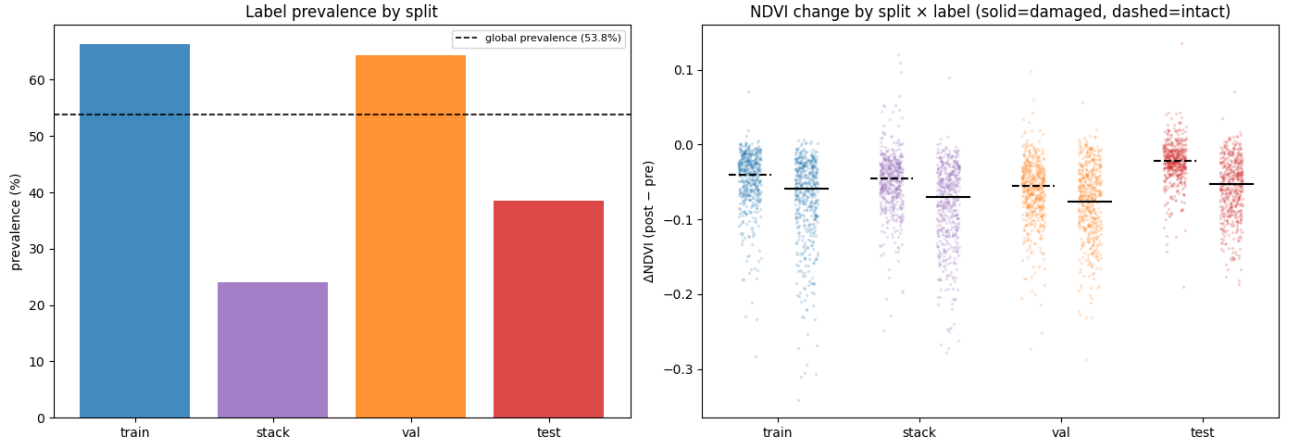


Figure A.6: Label prevalence by split against the global rate (left), and ΔNDVI by split and binary label, damaged vs. intact (right).

Table A.1: Sentinel-1 acquisition-date alignment for each city’s UNOSAT assessment dates and their temporal-stacking offsets (± 12 , ± 24 days). “Acquisition offset” is the calendar-day difference between the nearest available Sentinel-1 pass (Sentinel-1 runs in a 12-day orbit) and the desired date (positive: after; negative: before; desired date = UNOSAT date + temporal offset). Chernihiv and Rubizhne, the +12d and +24d offsets resolve to the same acquisition, so the +24d composite duplicates +12d rather than adding independent temporal information.

City (UNOSAT date)	Offset (d)	Nearest S1 acquisition	Δ (d)
Gaza (2024-05-03)	-24	2024-04-10	+1
	-12	2024-04-22	+1
	0	2024-05-04	+1
	+12	2024-05-16	+1
	+24	2024-05-28	+1
Gaza (2024-07-06)	-24	2024-06-09	-3
	-12	2024-06-21	-3
	0	2024-07-03	-3
	+12	2024-07-15	-3

continued on next page

City (UNOSAT date)	Offset (d)	Nearest S1 acquisition	Δ (d)
Gaza (2024-09-06)	+24	2024-07-27	-3
	-24	2024-08-08	-5
	-12	2024-08-20	-5
	0	2024-09-01	-5
	+12	2024-09-13	-5
	+24	2024-09-25	-5
Raqqa (2017-10-21)	-24	2017-09-27	0
	-12	2017-10-09	0
	0	2017-10-21	0
	+12	2017-11-02	0
	+24	2017-11-14	0
Mosul (2017-08-04)	-24	2017-07-08	-3
	-12	2017-07-20	-3
	0	2017-08-01	-3
	+12	2017-08-13	-3
	+24	2017-08-25	-3
Chernihiv (2022-04-28)	-24	2022-04-08	+4
	-12	2022-04-20	+4
	0	2022-05-02	+4
	+12	2022-05-14	+4
	+24	2022-05-14	-8
Rubizhne (2022-07-09)	-24	2022-06-20	+5
	-12	2022-07-02	+5
	0	2022-07-14	+5
	+12	2022-07-26	+5
	+24	2022-07-26	-7

B Training

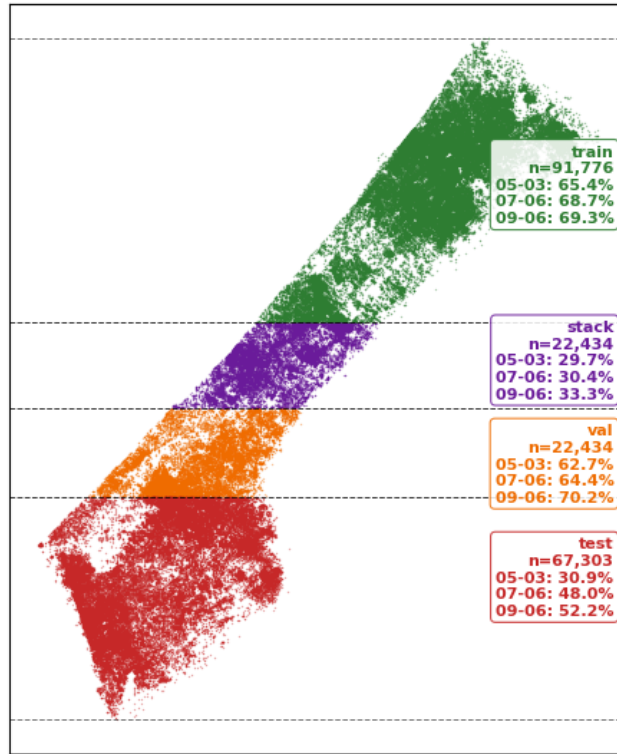
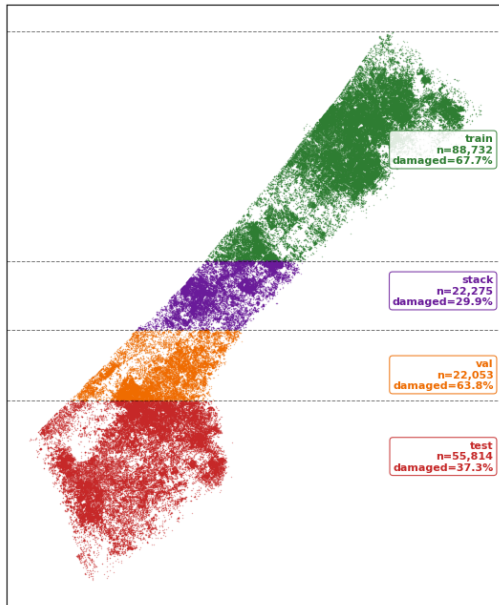
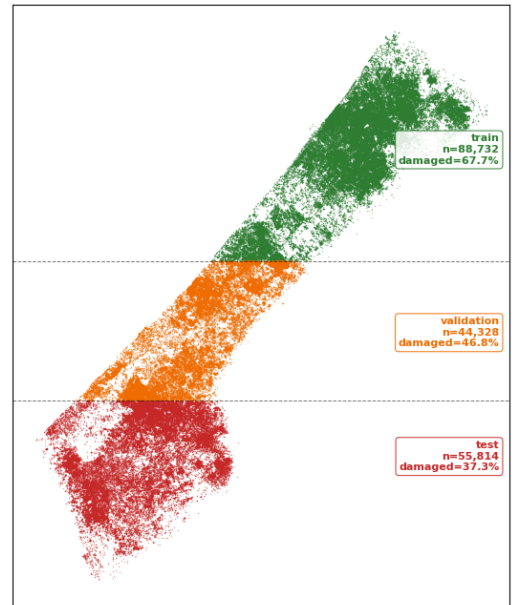


Figure B.1: Latitude-based spatial split for the SAR experiments. Labels indicate the number of unique building footprints per band and damage prevalence.



(a) From-Scratch



(b) Pretrained

Figure B.2: Latitude-based spatial splits for the Planet experiments. Labels indicate the number of building footprints and damage prevalence in each band.

Table B.1: Optuna hyperparameter search space. Both datasets use `TPESampler` (seeded), 10 trials, 25 max epochs per trial, early-stopping patience 6, trained on a 50% subsample of the training band, selecting on validation ROC AUC. `head_dim/head_dropout` apply only to the Siamese architecture.

Hyperparameter	Distribution	SAR range	Planet range
Learning rate	log-uniform	$[10^{-4}, 3 \times 10^{-3}]$	$[10^{-4}, 3 \times 10^{-3}]$
Weight decay	log-uniform	$[10^{-6}, 10^{-2}]$	$[10^{-6}, 10^{-2}]$
Batch size	categorical	$\{64, 128, 256\}$	$\{64, 128, 256\}$
Dropout	uniform	$[0.1, 0.5]$	$[0.1, 0.5]$
Width	categorical	$\{8, 16, 32\}$	$\{8, 16, 32\}$
Depth	integer	$[2, 4]$	$[2, 5]$
Head dim. (Siamese)	categorical	$\{64, 128, 256\}$	$\{64, 128, 256\}$
Head dropout (Siamese)	uniform	fixed at 0.2 (not tuned)	$[0.1, 0.4]$

Planet’s depth upper bound is derived from patch size ($\lfloor \log_2(32) \rfloor = 5$) rather than hardcoded as in the SAR notebook. Planet’s 3m resolution (vs. Sentinel-1’s 10m) resolves more spatial detail per patch, motivating the deeper search space and, correspondingly, the additional head-dropout tuning to regularize the larger resulting models against overfitting.

C Evaluation

Table C.1: Model size, Sentinel-1 SAR dataset.

Architecture	Setting	Parameters
Base CNN	default	1,529
Base CNN	HPO-tuned	19,937
Siamese CNN	default	7,769
Siamese CNN	HPO-tuned	118,113

Table C.2: XGBoost stacker complexity, Sentinel-1 SAR dataset. Trees and leaves reflect the effective ensemble actually used at prediction time (`best_iteration`+1 trees).

CNN run	Stacker variant	Trees	Leaves	Features
<i>Base CNN</i>				
Baseline	spatial	38	608	5
Baseline	spatial + temporal	49	783	11
HPO	spatial	75	1,193	5
HPO	spatial + temporal	50	800	11
<i>Siamese CNN</i>				
Baseline	spatial	35	560	5
Baseline	spatial + temporal	77	1,227	11
HPO	spatial	19	304	5
HPO	spatial + temporal	49	784	11

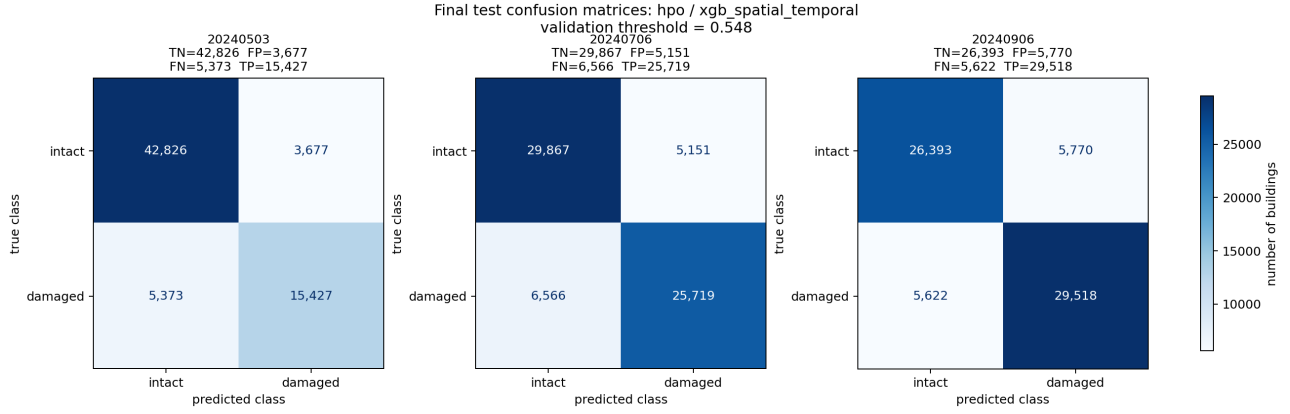


Figure C.1: Confusion matrices for the final Base CNN pipeline (HPO CNN + spatial-temporal XGBoost stacker) on the sealed Sentinel-1 SAR test set, one panel per assessment date, at the validation-selected threshold (0.548).

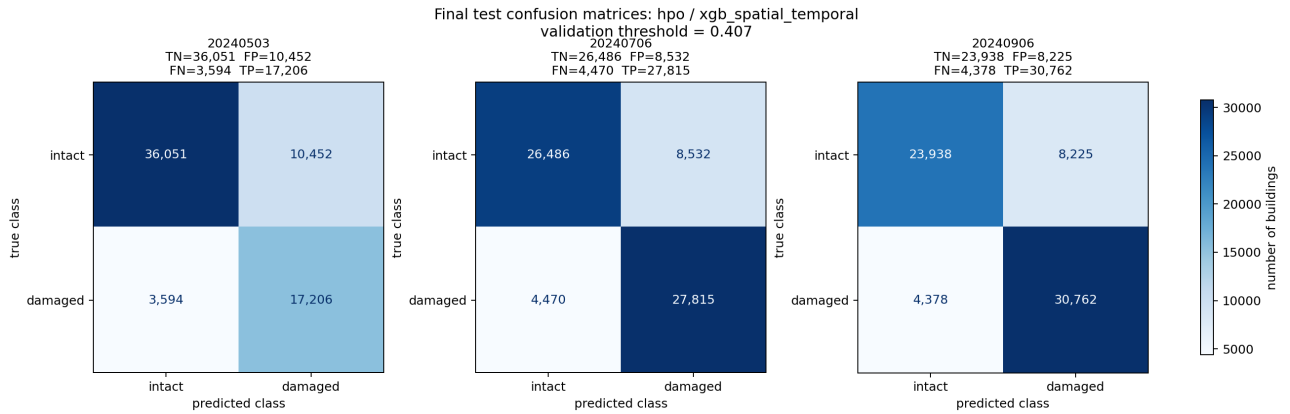


Figure C.2: Confusion matrices for the final Siamese CNN pipeline (HPO CNN + spatial-temporal XGBoost stacker) on the sealed Sentinel-1 SAR test set, one panel per assessment date, at the validation-selected threshold (0.407).

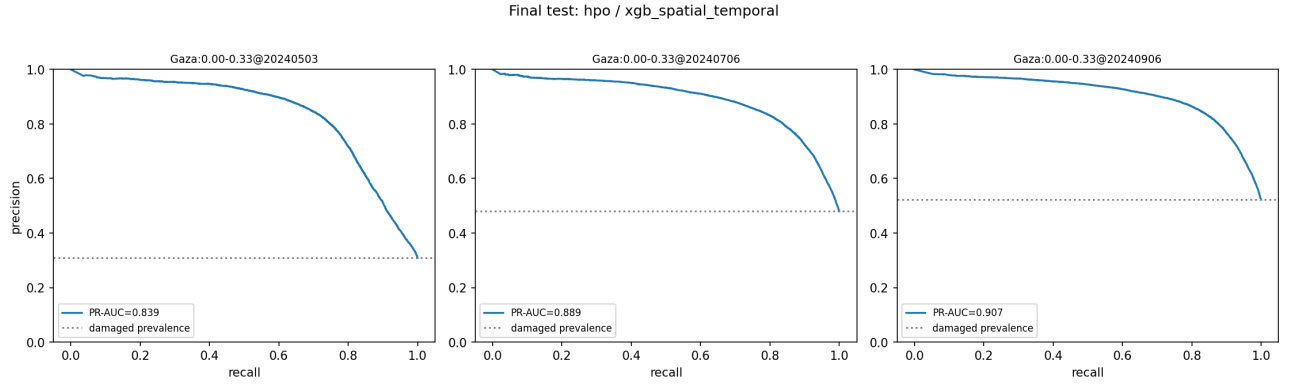


Figure C.3: Precision-recall curves for the final Base CNN pipeline on the sealed Sentinel-1 SAR test set, one panel per assessment date. The dashed horizontal line marks that date's damage prevalence.

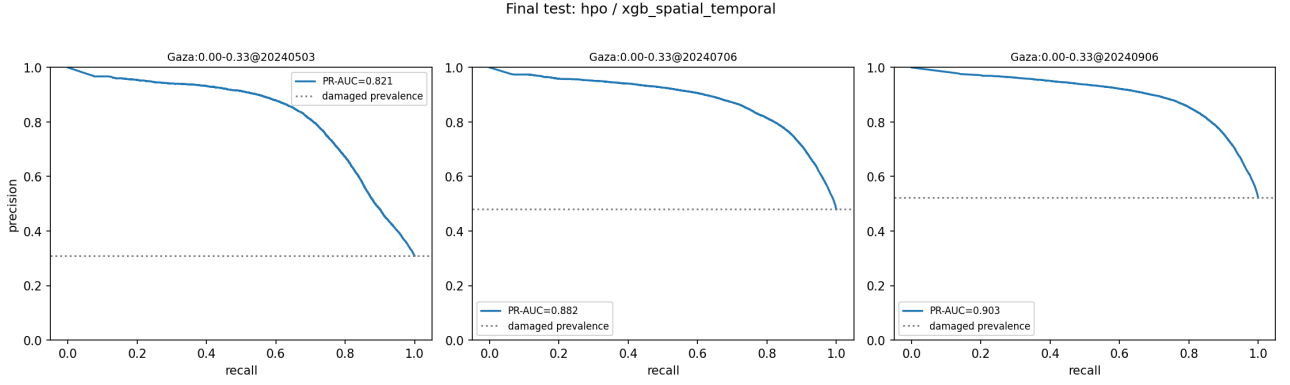


Figure C.4: Precision-recall curves for the final Siamese CNN pipeline on the sealed Sentinel-1 SAR test set, one panel per assessment date. The dashed horizontal line marks that date's damage prevalence.

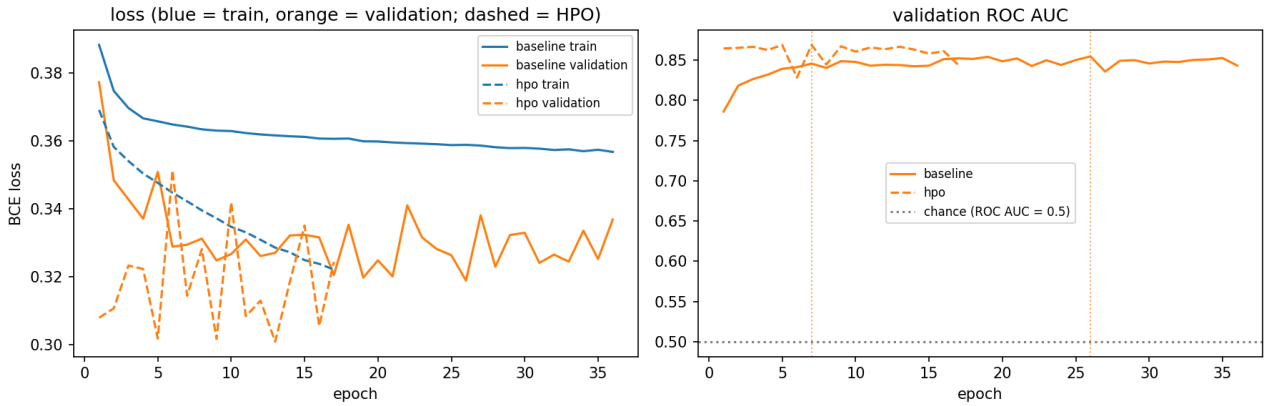


Figure C.5: Training curves for the Base CNN architecture on Sentinel-1 SAR data: binary cross-entropy loss (left) and validation ROC AUC (right) by epoch, comparing hand-set (baseline, solid) and Optuna-tuned (HPO, dashed) hyperparameters.

Table C.3: Generalization of the selected Base CNN pipeline to four unseen cities: Raqqa (2017-10-21), Mosul (2017-08-04), Chernihiv (2022-04-28), and Rubizhne (2022-07-09). None were used for training, stacking, validation, or threshold selection. $F1_{\text{oracle}}$ is the best-possible F1 at the optimal threshold for that city (computed using test labels, i.e. an upper bound, not what the frozen pipeline achieves).

City	n	Prevalence	Precision	Recall	F1	$F1_{\text{oracle}}$	ROC AUC
Raqqa	26,981	41.1%	0.650	0.657	0.654	0.670	0.767
Mosul	151,773	10.5%	0.258	0.726	0.381	0.515	0.813
Chernihiv	32,541	2.7%	0.414	0.052	0.093	0.250	0.754
Rubizhne	10,984	25.4%	0.632	0.403	0.492	0.550	0.762

Table C.4: Model size, Planet dataset. Parameter counts for Base/Siamese CNN verified directly from instantiated models; RSP ResNet-50 verified from the saved checkpoint’s `state_dict`.

Architecture	Setting	Parameters
Base CNN	default	1,817
Base CNN	HPO-tuned	98,769
Siamese CNN	default	7,913
Siamese CNN	HPO-tuned	196,881
RSP ResNet-50	pretrained	25,081,409

Table C.5: XGBoost stacker complexity, Planet dataset (spatial features only; no temporal offsets available). Trees and leaves reflect the effective ensemble at prediction time (`best_iteration+1` trees).

Architecture	CNN run	Trees	Leaves	Features
Base CNN	Baseline	8	128	5
Base CNN	HPO	49	766	5
Siamese CNN	Baseline	12	191	5
Siamese CNN	HPO	46	729	5

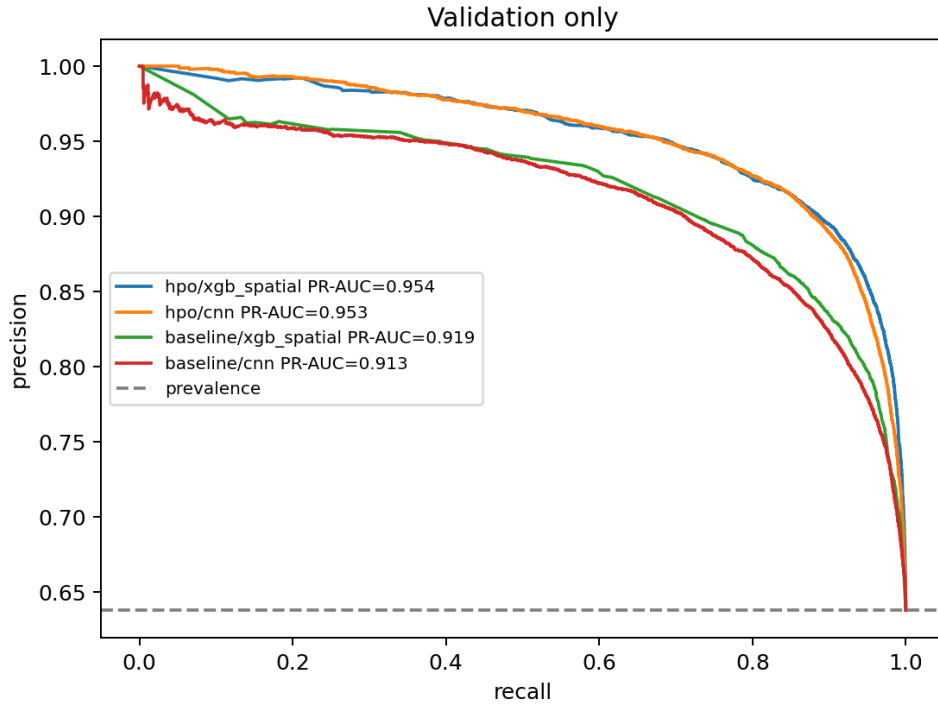


Figure C.6: Validation precision-recall curves for the Base CNN, comparing baseline and HPO CNN scores with and without the spatial XGBoost stacker.

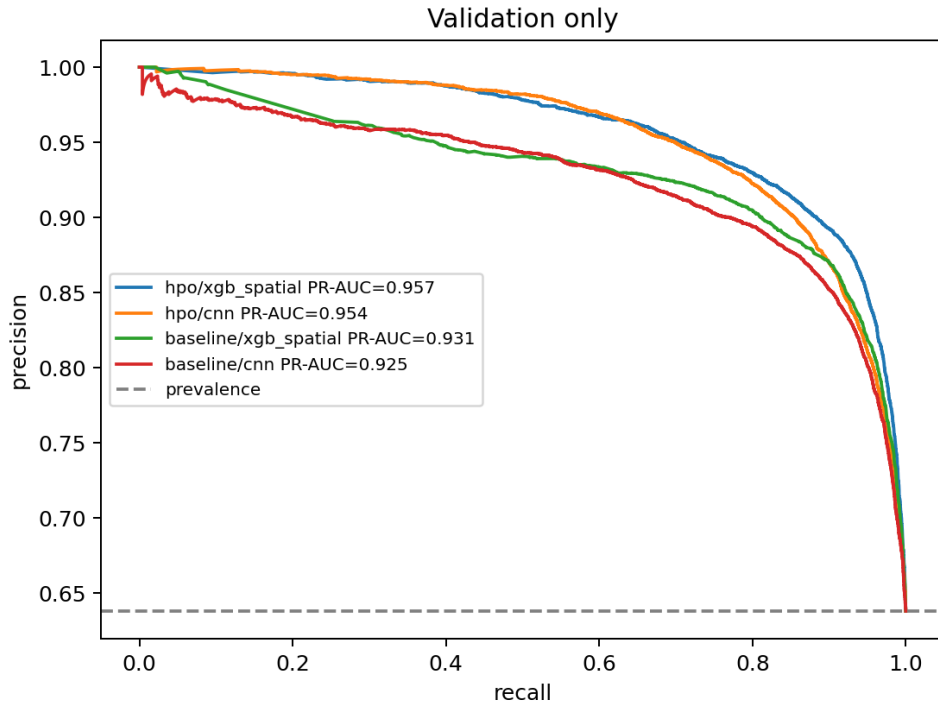


Figure C.7: Validation precision-recall curves for the Siamese CNN, comparing baseline and HPO CNN scores with and without the spatial XGBoost stacker.

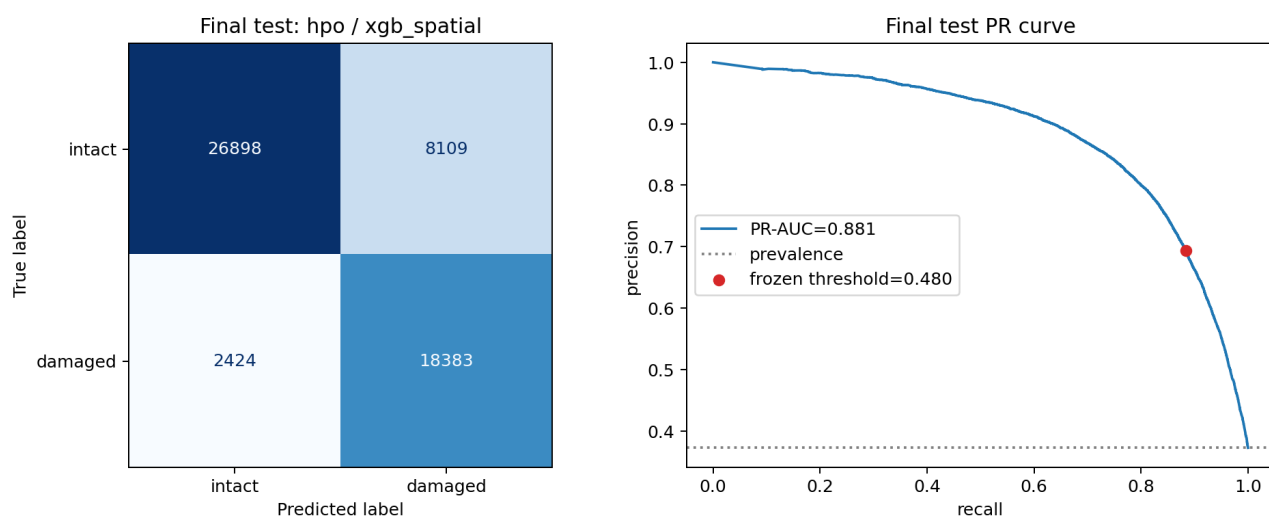


Figure C.8: Final test confusion matrix and precision-recall curve for the Base CNN's selected pipeline (HPO CNN with spatial XGBoost stacker), at the frozen threshold of 0.480.

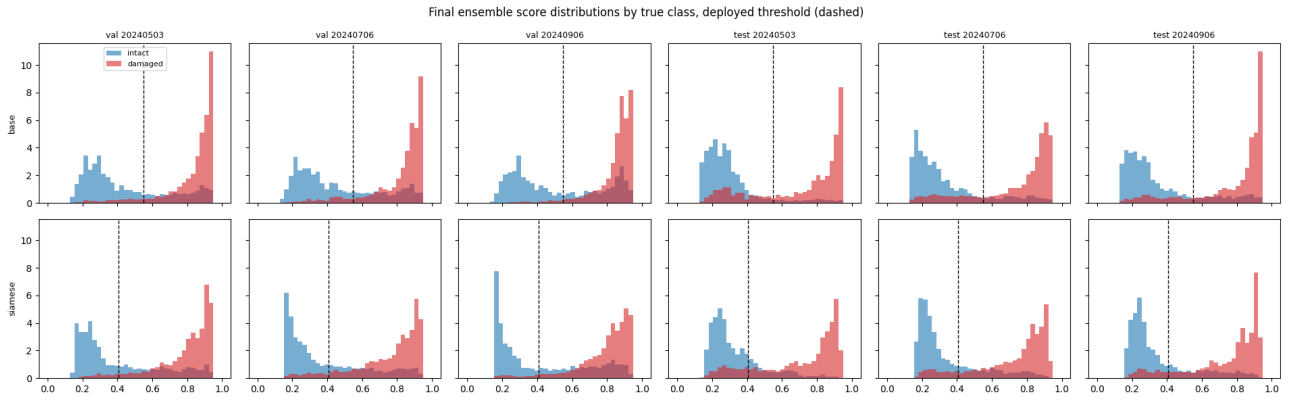


Figure C.9: Final ensemble score distributions (CNN plus spatial-temporal XGBoost stacker) by true class for SAR data, Base (top) and Siamese (bottom), for every validation and test date. The dashed line marks each model's deployed decision threshold (0.548 for Base, 0.407 for Siamese).

Planet: final ensemble score distributions by true class, deployed threshold (dashed)

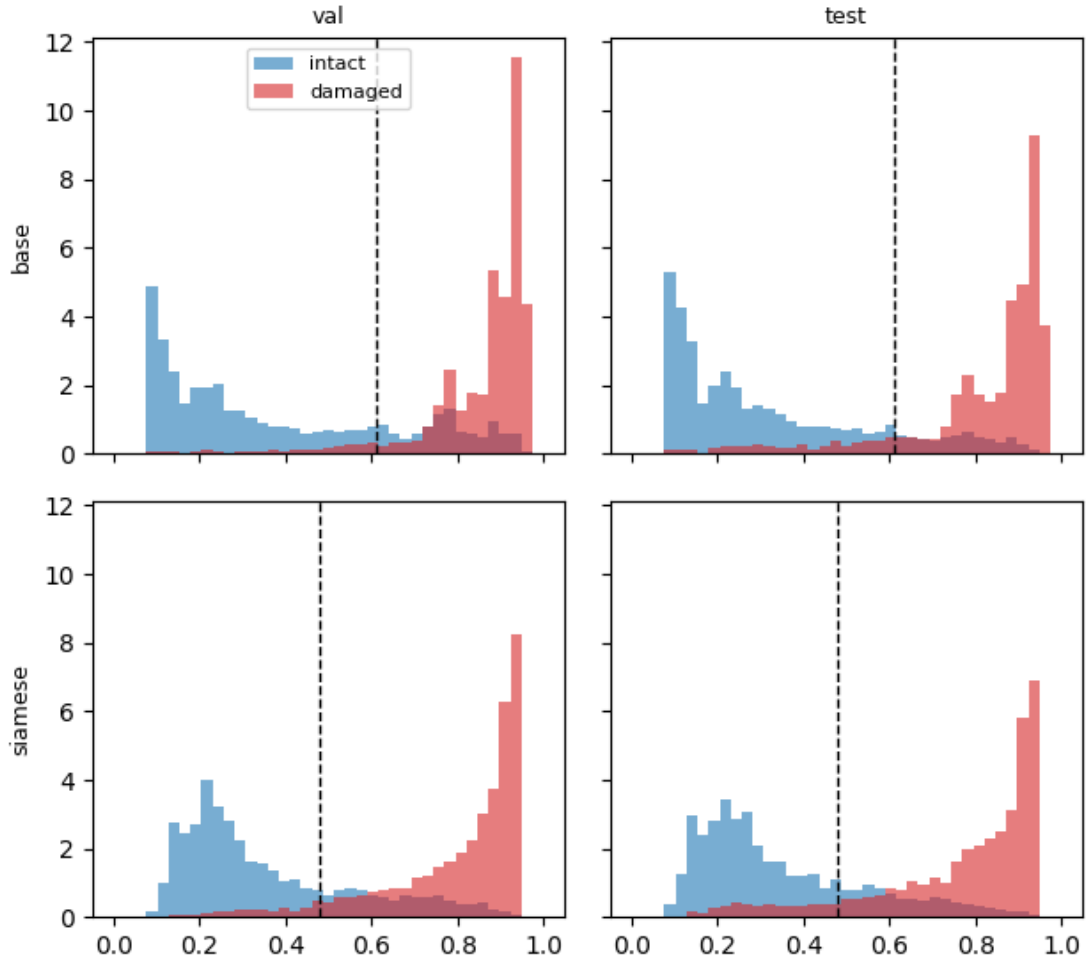


Figure C.10: Final ensemble score distributions (CNN plus spatial XGBoost stacker) by true class for Planet data, Base (top) and Siamese (bottom), for the validation and test bands. The dashed line marks each model's deployed decision threshold (0.613 for Base, 0.480 for Siamese).



Figure C.11: Planet Base CNN predictions overlaid on Sentinel 2 imagery. Color intensity denotes model confidences in deciles.